PTUT — FINAL YEAR PROJECT

Weekly Progress Report · Week 9 of 12

TaxWise Freelance

Automated Tax Deductions & Invoicing Platform

Admin Workstation & Consumer UX Polish

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DOMAIN
FinTech, Personal Wealth & Micro-Finance

Source code: github.com/SohaibRamzan01/taxwise-freelance

1. Summary

This week added a **protected admin portal** (only users with the ADMIN role can open it) with **analytics charts** and an **audit log**, and polished the experience for normal users: **mobile layout, toast messages, confirmations, loading and error pages**.

RBAC
admin role checked on the server for every admin request

3
weekly analytics charts + totals

320 px+
works on phone screens with no sideways scrolling

Task from the plan	Status
Protected Admin Portal with Role-Based Access Control	Done
Telemetry charts and governance analytics	Done — 3 weekly charts, platform totals, audit log viewer
Responsive mobile layouts	Done — mobile menu, tables become cards
Toast notifications	Done — success and error toasts for every change

2. Admin portal and role-based access control

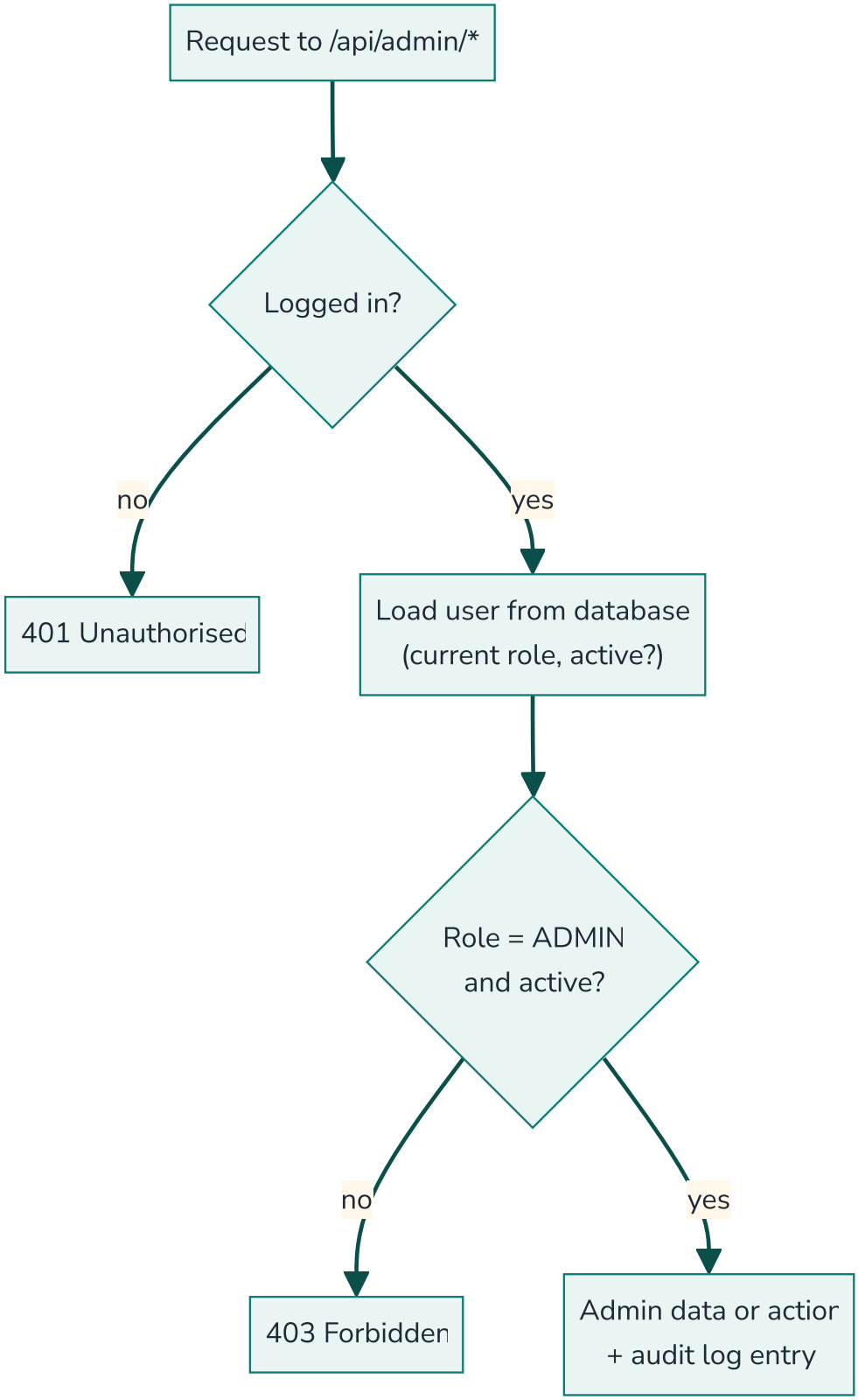


Figure 1 — Every admin request checks the user's current role in the database

- **Server-side checks:** every admin API checks the role **from the database**, not just from the login token, so a role change takes effect immediately.

- **Users page:** search and page through users, change a user's role, enable or disable an account.
- **Safety rule:** an admin cannot demote or disable **themselves** (prevents locking everyone out).
- **Audit log viewer:** shows logins, deletes, bulk actions, role changes and background job runs, with filters.

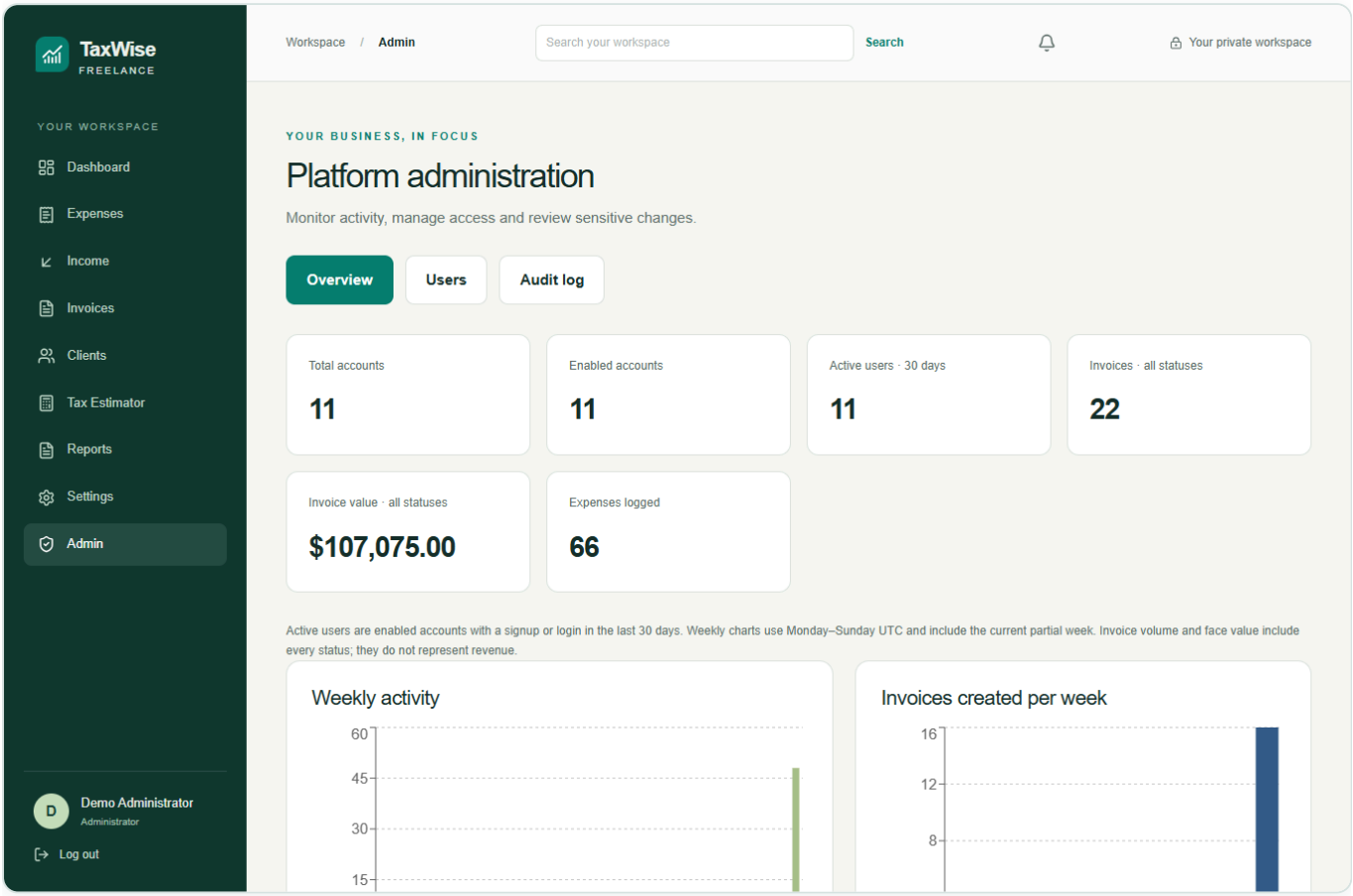


Figure 2 — Admin overview: platform totals and weekly activity charts

3. Analytics (telemetry) charts

Chart	What it shows
Weekly activity	Sign-ups, active users and expenses logged per week (last 12 weeks)
Invoices created per week	How much invoicing is happening
Invoice value per week	Total value of invoices created
Platform totals	Accounts, active users (30 days), invoices, expenses

Each chart also has a **data table** for screen-reader users.

4. User experience polish

Improvement	Details
Mobile layout	Menu button on small screens; wide tables turn into easy-to-read cards; checked with no sideways scrolling at 390 px
Toast messages	A short message after every save, delete, import or error
Confirm dialogs	Deleting, voiding or un-paying asks "are you sure?" first
Loading & error states	Skeleton loaders, friendly 404 and error pages, empty-state messages
Accessibility	Labels on all inputs, visible keyboard focus, "skip to content" link
Clear wording	Tax figures use the same labels on every page with a one-line explanation

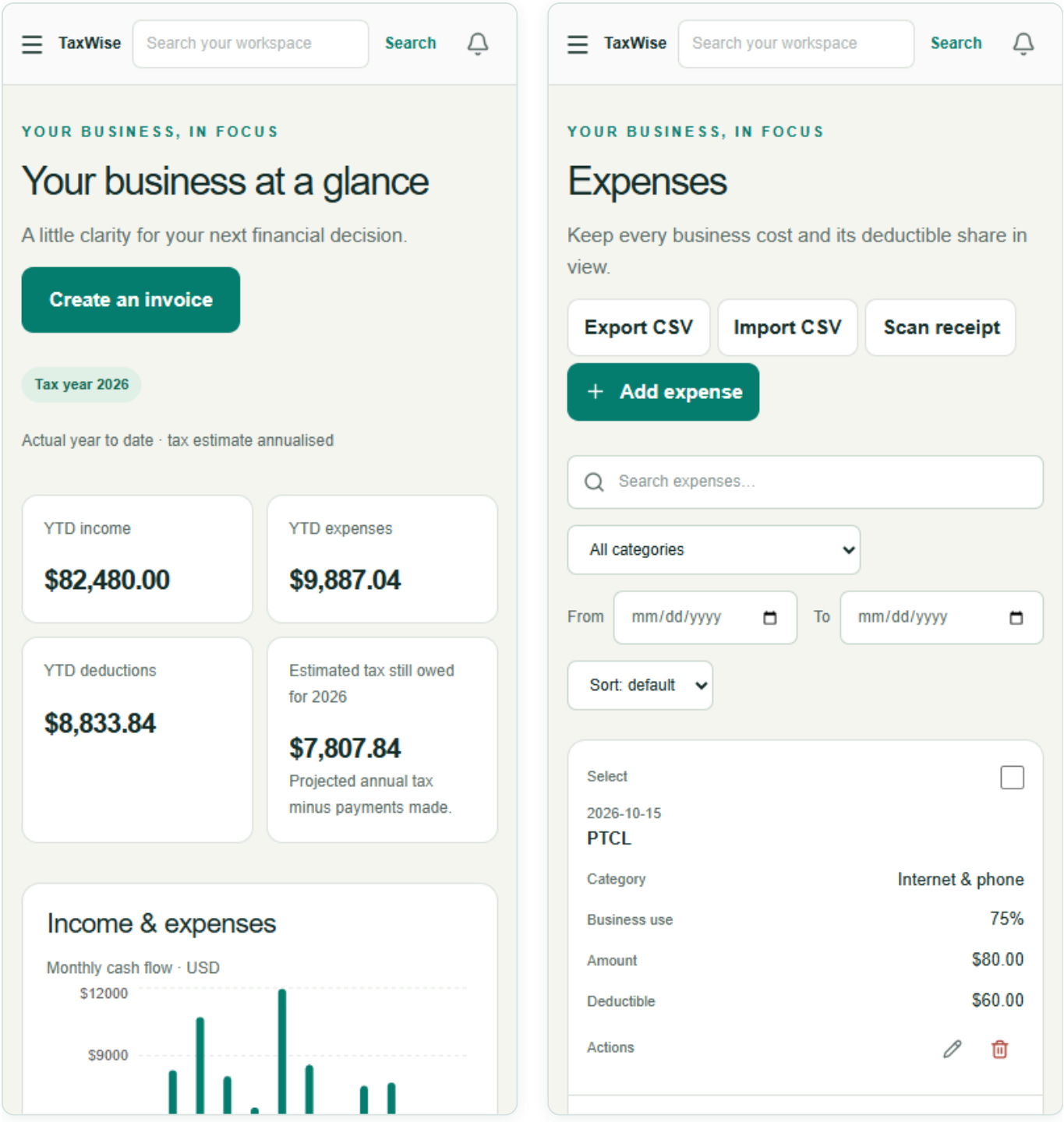


Figure 3 — Dashboard (left) and Expenses as cards (right) on a phone-sized screen

5. Visual QA

All 13 screens were opened in a real browser at desktop (1366 px) and phone (390 px) size and checked for layout problems and console errors. Issues found and fixed: the sidebar stopping half-way down long pages, a missing "deductible" column on expenses, a confusing quarterly status, a missing site icon, and two tax figures that looked contradictory.

6. Testing and verification

Check	Result
Admin access tests	Normal users get 403 on every admin route; disabled or demoted admins lose access immediately
Self-protection tests	Admin cannot demote or disable their own account
Visual QA	13 screens × 2 sizes — all load, no console errors, no horizontal overflow
Automated tests	All passing; production build passing

7. Challenges and solutions

Challenge	Solution
A login token could still say "admin" after the role was removed	Admin checks read the current role from the database on every request
Charts are hard for screen-reader users	Each chart has a matching data table
Wide tables do not fit on phones	Tables switch to a card layout below tablet width

8. Plan for next week (Week 10)

- Raise and enforce test coverage on the core logic (target ≥ 70%).
- Run static security scans and fix dependency vulnerabilities.
- Review every input against the OWASP Top 10 risks.